

Statistical Analysis of LLM Evaluation Quality Patterns

Project Overview

This project demonstrates a comprehensive approach to analyzing quality and consistency in large-scale language model evaluation systems. Using synthetic data that mirrors real-world evaluation patterns, I built an end-to-end analytical pipeline to assess rater reliability, accuracy trends, and systematic evaluation patterns.

Business Context

Large language model evaluation relies on human raters to assess response quality, accuracy, and appropriateness. Understanding evaluation quality requires analyzing:

- **Inter-rater reliability:** Do raters agree on quality assessments?
- **Accuracy patterns:** How do rater judgments compare to ground truth?
- **Systematic biases:** Are certain evaluation types more prone to disagreement?
- **Quality drivers:** What factors predict evaluation difficulty?

Technical Approach

Data Generation

Created synthetic evaluation dataset (1,000 evaluations, 15 raters) with realistic patterns:

- Multiple raters per evaluation (mimicking real annotation workflows)
- Varying evaluation characteristics (complexity, domain, response length)
- Ground truth labels with realistic rater noise
- Systematic disagreement patterns based on evaluation difficulty

Analysis Pipeline

1. **Data Preparation:** Structuring evaluation data for analysis
2. **Exploratory Analysis:** Initial pattern discovery
3. **Inter-Rater Reliability:** Statistical agreement metrics (Fleiss' Kappa, Cohen's Kappa)
4. **Accuracy Analysis:** Performance metrics and systematic bias detection
5. **Visualization:** Publication-quality plots for stakeholder communication

Technologies Used

- **R** (tidyverse, ggplot2, irr)
- **Statistical Methods:** Inter-rater reliability, correlation analysis, hypothesis testing
- **Reproducible Research:** R Markdown for automated reporting

Key Findings

[To be completed after analysis]

Inter-Rater Agreement

- Overall agreement rate: XX%
- Fleiss' Kappa: X.XX (interpretation)
- Agreement varies by evaluation type: [pattern description]

Accuracy Patterns

- Mean accuracy vs ground truth: XX%
- Systematic biases identified in: [domains/types]
- Difficulty predictors: [characteristics]

Actionable Insights

- Rater training recommendations
- Evaluation guideline improvements
- Quality assurance focus areas

Repository Structure

```
portfolio-llm-eval-quality/
|
├── README.md           # This file
├── data/
|   ├── generate_synthetic_data.R  # Synthetic dataset creation
|   |
|   └── analysis/
|       ├── 01_data_preparation.R    # Data cleaning and structuring
|       ├── 02_exploratory_analysis.R # Initial pattern discovery
|       └── 03_inter_rater_reliability.R # IRR statistical analysis
```

```
| | — 04_accuracy_analysis.R      # Performance metrics
| | — 05_visualizations.R        # Publication-quality plots
|
| — output/
| | — figures/                  # Generated visualizations
| | — summary_report.html       # Complete analysis report
|
| — docs/
|   — methodology.md           # Technical methodology documentation
```

Running the Analysis

```
r

# 1. Generate synthetic data
source("data/generate_synthetic_data.R")

# 2. Run analysis pipeline
source("analysis/01_data_preparation.R")
source("analysis/02_exploratory_analysis.R")
source("analysis/03_inter_rater_reliability.R")
source("analysis/04_accuracy_analysis.R")
source("analysis/05_visualizations.R")

# 3. Generate report
rmarkdown::render("output/summary_report.Rmd")
```

Skills Demonstrated

- **Statistical Analysis:** Inter-rater reliability metrics, hypothesis testing, correlation analysis
- **Data Manipulation:** Complex data restructuring, aggregation, transformation
- **Data Visualization:** Professional plots for technical and executive audiences
- **Reproducible Research:** Documented, version-controlled analysis pipeline
- **Domain Expertise:** LLM evaluation quality assessment
- **Communication:** Translating technical findings to actionable insights

Future Enhancements

- Machine learning models to predict evaluation difficulty
- Natural language processing of rater comments/feedback

- Time-series analysis of rater performance trends
- Interactive dashboard for quality monitoring

About This Project

This portfolio project was developed to demonstrate data science capabilities in the context of large-scale evaluation systems. The synthetic data mimics realistic patterns observed in LLM evaluation workflows while maintaining complete independence from any proprietary systems.

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All data in this project is synthetically generated. No proprietary or confidential information is included.